Rendeiro 2020 CLL — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

7 samples / 4 patients (CLL1/5/6/8) intermix on one continuous manifold (iLISI 1.0→4.7 of 7; radius-graph: 166 detached cells stripped); one diffuse compartment, no separable cycling lobe.

Gate 2 — module geography → GO (non-branching)

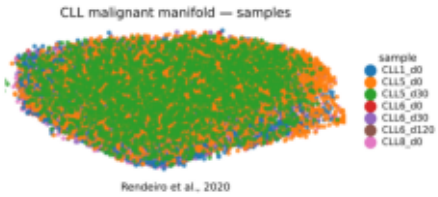
Gast declines / pSom rises on one axis (ρ=−0.62 / +0.49); Crest anti-correlated with both (raw r=−0.55 / −0.55), rising toward a terminal readout (crest-high decile mean pt=0.73).

Gate 3 — hub mutual exclusivity → NA

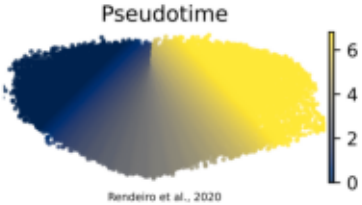
Single monotone axis — no bifurcation and no Mixed-type hub, so there is no branch pair to test for mutual cross-repression.

STEP FIGURES

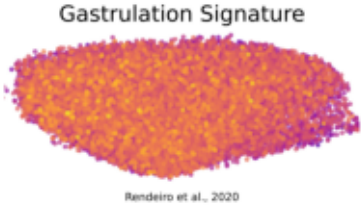
Gate 1: integration



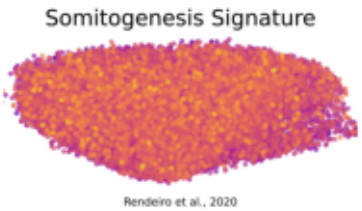
Trajectory / root



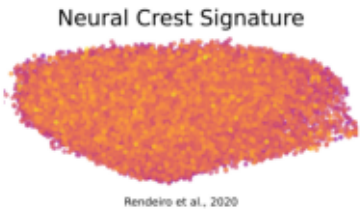
Gast



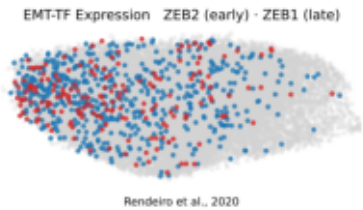
pSom



Crest



EMT-TFs



Dataset. Chronic Lymphocytic Leukemia (Rendeiro et al., 2020), 10x — peripheral blood, longitudinal ibrutinib cohort. Malignant only: 18,854 load; 18,657 across 7 samples / 4 patients (≥284, multiplicative gap 1012→80) enter preprocessing; 18,491 after island removal enter the trajectory (whole manifold, no lobe scoping — proliferation is diffuse, not a separable compartment).

Verdict. ✓ Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Rendeiro's CLL malignant manifold is a single monotone Gast → terminal-Crest gradient (Linear-pSom/Crest), matching the developmental axis; it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. A PPT tree fabricates 4–7 forks on this compact near-round cloud, while an ElPiGraph curve gives a clean fold-free 2-tip backbone (self-approach 2.80) — the manifold is genuinely non-branching, not a collapsed Y. EMT-TFs are near-silent in this blood cancer: only ZEB2 (12%) / ZEB1 (4%) clear the 3% detection floor (SNAI/TWIST/FOXC/PRRX essentially unexpressed), and their pseudotime trend is flat (ρ≈0), reported as-is.

Parameter choices

Min cells/sample	≥284 (keeps 7/11; gap 1012→80)	**Integration**	Harmony θ=5; 20 PC; spread=1.3
Scope	whole manifold (no scoping)	**Trajectory**	curve N=20 λ=0.3 μ=100
Root node	tip 2 = Gast/pSom-high, Crest-low	**Geometry**	Linear-pSom/Crest